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QUESTION: Write a menu-driven C program to perform the following operations using separate functions:

1. Count characters in a string without using built-in length functions.
2. Print each character of a string on a new line.
3. Count vowels and consonants in a string.
4. Exit.

SOLUTION:

1. Aim

To write a menu-driven C program using **separate functions** to perform the following string operations:

1. Count the number of characters in a string without using built-in length functions such as strlen().
2. Print each character of the string on a new line.
3. Count the number of vowels and consonants in the string.
4. Exit the program.

2. Algorithm

Main Algorithm

1. Start the program.
2. Input a string from the user.
3. Display the following menu:
    - Count characters in the string.
    - Print each character on a new line.
    - Count vowels and consonants.
    - Exit.
4. Accept the user's choice.
5. Use a switch statement to call the corresponding function.

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6. Repeat the menu until the user selects **Exit**.
7. Stop.

Operation 1: Count Characters

1. Initialize a variable count = 0.
2. Start from the first character of the string.
3. Traverse the string until the null character '\0' is encountered.
4. Increment count for every character.
5. Display the value of count.

Note: No built-in length function such as strlen() is used.

Operation 2: Print Each Character on a New Line

1. Start from the first character of the string.
2. Traverse the string until '\0' is encountered.
3. Print the current character.
4. Move to the next line.
5. Continue until all characters are printed.

Operation 3: Count Vowels and Consonants

1. Initialize vowels = 0 and consonants = 0.
2. Traverse the string character by character.
3. Check whether the character is an alphabet.
4. If it is a, e, i, o, u or their uppercase equivalents, increment vowels.
5. Otherwise, if it is an alphabet, increment consonants.
6. Ignore spaces, digits and special characters.
7. Display the number of vowels and consonants.

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C PROGRAM:

```
#include <stdio.h>

// Function declarations
void countCharacters(char str[]);
void printCharacters(char str[]);
void countVowelsConsonants(char str[]);

int main()
{
   char str[100];
   int choice;

   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);

   // Remove newline character added by fgets()
   int i = 0;

   while(str[i] != '\0')
   {
      if(str[i] == '\n')
      {
         str[i] = '\0';
         break;
      }
```

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```
    i++;
}

do
{
    printf("\n----- STRING OPERATIONS MENU -----\n");
    printf("1. Count characters in the string\n");
    printf("2. Print each character on a new line\n");
    printf("3. Count vowels and consonants\n");
    printf("4. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            countCharacters(str);
            break;

        case 2:
            printCharacters(str);
            break;

        case 3:
```

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```
            countVowelsConsonants(str);
            break;

        case 4:
            printf("Program exited successfully.\n");
            break;

        default:
            printf("Invalid choice! Please try again.\n");
      }

   } while(choice != 4);

   return 0;
}



// Function to count characters without strlen()
void countCharacters(char str[])
{
   int i = 0;
   int count = 0;

   while(str[i] != '\0')
   {
      count++;
```

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```
      i++;
   }

   printf("\nNumber of characters = %d\n", count);
}



// Function to print each character on a new line
void printCharacters(char str[])
{
   int i = 0;

   printf("\nCharacters of the string are:\n");

   while(str[i] != '\0')
   {
      printf("%c\n", str[i]);
      i++;
   }
}



// Function to count vowels and consonants
void countVowelsConsonants(char str[])
{
   int i = 0;
```

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```
int vowels = 0;
int consonants = 0;
char ch;

while(str[i] != '\0')
{
   ch = str[i];

   // Check whether character is an alphabet
   if((ch >= 'A' && ch <= 'Z') ||
     (ch >= 'a' && ch <= 'z'))
   {
      // Check for vowels
      if(ch == 'A' || ch == 'E' || ch == 'I' ||
         ch == 'O' || ch == 'U' ||
         ch == 'a' || ch == 'e' || ch == 'i' ||
         ch == 'o' || ch == 'u')
      {
         vowels++;
      }
      else
      {
         consonants++;
      }
   }
```

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```
      i++;
    }

    printf("\nNumber of vowels = %d\n", vowels);
    printf("Number of consonants = %d\n", consonants);
}
```

TEST CASE TABLE:

Test Case	Input String	Operation	Expected Result
1	Hello	Count characters	5
2	Hello	Print characters	H, e, l, l, o on separate lines
3	Hello	Vowels & consonants	Vowels = 2, Consonants = 3
4	AEIOU	Vowels & consonants	Vowels = 5, Consonants = 0
5	Computer Science	Vowels & consonants	Vowels = 6, Consonants = 9
6	C123	Vowels & consonants	Vowels = 0, Consonants = 1
7	Any string	Exit	Program terminates
8	Any string	Invalid menu choice	Displays invalid choice message

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COMPILATION AND EXECUTION:



